METRIC
MIL-M-24731(NAVY)
31 August 1989

MILITARY SPECIFICATION

MULTIPLEXERS, DEMULTIPLEXERS, MULTIPLEXERS/DEMULTIPLEXERS (MULDEMS), FREQUENCY-DIVISION, FIBER OPTIC INTERFACEABLE, SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fiber optic interfaceable, frequency-division multiplexer and demultiplexer units (see 6.5.10 and 6.5.8) intended for Naval shipboard applications. The unit includes multiplexers, demultiplexers, and various fiber optic components. Electrical signals are input to and output from the unit; signals are transferred optically between the multiplexers and demultiplexers.

1.2 Classification. The units are classified as specified in 1.2.1 through 1.2.6 (see 3.1).

1.2.1 Equipment function. The equipment function designation defines the unit as a FDM, FDDM, or a MULDEM.

M - Frequency-division multiplexer (FDM).
D - Frequency-division demultiplexer (FDDM).
C - Frequency-division multiplexer/demultiplexer (MULDEM).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Multiplexed signal type. The multiplexed signal (see 6.5.22) type designation defines the multiplexed signal output of the FDM, the multiplexed signal input to the FDDM, or the multiplexed signal output and input, respectively, of the MULDEM as electrical or optical.

A - Electrical (for FDM and FDDM only).
B - Optical (for FDM and FDDM only).
C - Electrical and electrical (for MULDEM only).
D - Electrical and optical (for MULDEM only).
E - Optical and electrical (for MULDEM only).
F - Optical and optical (for MULDEM only).

1.2.3 Channel configuration. The channel configuration designation defines the number of input or output channels (subchannels) (see 6.5.5) supported by the FDM or FDDM respectively. The frequency-division MULDEM supports the input and output channel configuration in a symmetrical fashion.

A - One to 16 channels (subchannels).
B - Greater than 16 channels (subchannels).

1.2.4 Analog subchannel class. The analog subchannel class designation defines the highest required analog signal (see 6.5.1) subchannel bandwidth (see 6.5.2) at half-power points.

A - Audio/voice, not greater than 3 kilohertz (kHz).
B - Broadband audio, not greater than 25 kHz.
C - Control signaling, not greater than 300 kHz.
D - Low resolution video, not greater than 7 megahertz (MHz).
E - High resolution video, not greater than 20 MHz.
F - Low resolution sensor video, not greater than 30 MHz.
G - Medium resolution sensor video, not greater than 700 MHz.
H - High resolution sensor video, not greater than 10 gigahertz (GHz).
I - Specification sheet determined.
J - Not applicable.

1.2.5 Digital subchannel class. The digital subchannel class designation defines the highest equivalent digital data rate using nonreturn-to-zero (NRZ) (see 6.5.24) code.

A - Not greater than 1 kilobits per second (kbps).
B - Not greater than 19.2 kbps.
C - Not greater than 64 kbps.
D - Not greater than 100 kbps.
E - Not greater than 1 megabits per second (Mbps).
F - Not greater than 10 Mbps.
G - Not greater than 100 Mbps.
H - Not greater than 1 gigabit per second (Gbps).
I - Not greater than 10 Gbps.
J - Specification sheet (see 3.1) determined.
K - Not applicable.

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1.2.6 Style. The style designation defines the package configuration.

A - Microcircuit: integrated circuit (IC) or hybrid, package and mounting technique.
B - Stand-alone module.
C - Vertical rack plug-in module (including groups of circuit card assemblies).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-I-735	- Isopropyl Alcohol.

MILITARY

MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-E-16400	- Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for.
MIL-F-16884	- Fuel, Naval Distillate.
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-S-19500	- Semiconductor Devices, General Specification for.
MIL-E-21981	- Electronics Equipment, Nomenclature, Serial Numbers and Identification Plates: Requirements for.

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MILITARY (Continued)

MIL-I-23053 - Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for.
MIL-I-23053/6 - Insulation Sleeving, Electrical, Heat Shrinkable, Polyolefin, Semi-Rigid, Crosslinked.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-M-24731/1 - Multiplexers, Demultiplexers, Multiplexers/ Demultiplexers (MULDEMS), Multiplexer, Frequency-Division, Fiber Optic, Shipboard, 4:1 Channels, 30 MHz (Double Sideband) (Metric).
MIL-M-24731/2 - Multiplexers, Demultiplexers, Multiplexers/ Demultiplexers (MULDEMS), Demultiplexer, Frequency-Division, Fiber Optic, Shipboard, 1:4 Channels, 30 MHz (Double Sideband) (Metric).
MIL-T-24735 - Transmitters, Light Signal, Analog, Fiber Optic, Shipboard (Metric) General Specification for.
MIL-R-24737 - Receivers, Analog, Fiber Optic, Shipboard, General Specification for.
MIL-C-28754 - Connectors, Electrical, Modular, and Component Parts General Specification for.
MIL-C-28859 - Connector Component Parts, Electrical Backplane, Printed Wiring, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for.
MIL-P-50884 - Printed-Wiring, Flexible and Rigid-Flex.
MIL-T-55164 - Terminal Boards, Molded, Barrier Screw and Stud Types, and Associated Accessories, General Specification for.
MIL-C-85045 - Cable, Fiber Optic, Shipboard (Metric) General Specification for.
MIL-C-85045/1 - Cable, Fiber Optic, Fiber Optic Cable Configuration Type B (Pigtail), Optical Fiber Type MM (Graded Index, Glass Core and Glass Cladding, Multimode) (Metric).

STANDARDS

MILITARY

MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-449 - Radio Frequency Spectrum Characteristics Measurement of.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-965 - Parts Control Program.
MIL-STD-1344 - Test Methods for Electrical Connectors.
MIL-STD-1378 - Requirements for Employing Standard Electronic Modules.
MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1399, Section 071 - Interface Standard for Shipboard Systems Mass/Size/Shape, Shipboard Units. (Metric)
MIL-STD-1399, Section 300 - Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric)
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
DOD-STD-2167 - Defense System Software Development.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-241 - Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies.
MIL-HDBK-472 - Maintainability Prediction.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

F 1166 - Standard Practice for Human Engineering Designs for Marine Systems, Equipment and Facilities.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455-6 - Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices. (DoD adopted)
455-12 - Fluid Immersion Test Procedure for Fiber Optic Connecting Devices.

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EIA (Continued)
455-36 - Twist Test for Fiber Optic Connecting Devices. (DoD adopted)
455-118 - Output Propagation Delay.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Reliability. Reliability of the unit furnished under this specification shall be as specified (see 3.1) and maintained in accordance with the procedures and requirements specified in MIL-STD-790 and MIL-STD-690. The quantitative reliability requirements expressed in mean-time-between-failures (MTBF) (see 6.5.19), mean-cycle-between-failures, or others, shall be as specified (see 3.1).

3.3.1 Reliability prediction. A qualitative reliability prediction for the unit, in the form of a MTBF calculation, shall be performed in accordance with computational techniques specified in MIL-HDBK-217. Failure rates shall be derived based on an application environment assuming "Naval, sheltered" usage and an ambient thermal environment temperature range as specified herein (see 3.9).

3.3.2 Burn-in (style A units) (see 4.3.1.2). All units shall meet the requirements of MIL-STD-883, method 1015.

3.4 Materials. The units shall be constructed of materials that will not produce toxic, corrosive, or explosive by-products. Materials shall not have adverse effects upon operational or maintenance personnel under all operational and environmental conditions, nor cause degradation of equipment performance.

3.4.1 Fragile or brittle materials. Cast iron, ebonite (hard vulcanized rubber), asbestos, porcelain, and other similar materials shall not be used.

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3.4.2 Hazardous material marking. Units produced with hazardous materials, only as required (see 3.1), shall be marked in accordance with the requirements of public law and regulations. The marking shall include as applicable: name of product; quantity; warning symbol; signal word designating degree of hazard; affirmative statement of hazards; precautionary measures covering actions to be followed or avoided; instructions in case of contact or exposure; antidotes and notes to physicians; instructions in case of fire, spill, or leak; instructions for handling and storage; and disposal instructions. Marking of hazardous materials shall be in conformance with Federal Hazardous Substance Labeling Act.

3.4.3 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification.

3.5 Design and construction (style A units only). For style A units only, the construction and physical dimensions shall be as specified in 3.5.1 through 3.5.6.

3.5.1 Case (see 4.6.2.1). The unit shall be hermetically sealed. Dimensions, pinouts, and pinout functions shall be as specified (see 3.1). Exclusive of the connectors, the dimensions of the unit shall not exceed the dimensions as specified (see 3.1). The unit case may be electrically grounded, connected to a voltage pin, or electrically isolated (that is, floating) as specified (see 3.1).

3.5.2 Terminals.

3.5.2.1 Terminal strength (see 4.6.2.2). The electrical terminals interface (see 6.5.18) shall reveal no evidence of breakage, loosening, or relative motion between the electrical terminals or connector, and body of the unit.

3.5.2.2 Solderability (see 4.6.2.3). Terminal solderability shall meet the temperature and time requirements as specified in MIL-STD-883, method 2003, or as specified (see 3.1).

3.5.3 Finish (see 4.6.1). The finish on style A units shall conform to the requirements specified in MIL-M-38510.

3.5.4 Electrostatic discharge sensitivity (ESDS) (see 4.6.2.4). Units shall meet the performance requirements specified herein and the requirements for ESDS in MIL-STD-883, class 2 devices.

3.5.5 Semiconductor devices. Semiconductor devices used in the unit shall be in accordance with MIL-S-19500.

3.5.6 Microcircuit devices. Microcircuit devices used in the unit shall be in accordance with MIL-M-38510.

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3.6 Design and construction (styles B and C units only). For styles B and C units, the construction and physical dimensions shall be as specified in 3.6.1 through 3.6.14. Unless otherwise specified (see 3.1), all units shall use modular construction. The equipment modular construction shall conform to the standard hardware program module requirements specified in MIL-STD-1378. The selection of specific assembly design and packaging techniques shall reflect the reliability and maintainability requirements as specified herein.

3.6.1 Degree of enclosure. Unless otherwise specified (see 3.1), the type of enclosure shall be either hermetic or watertight (see 3.5.1 and 3.9.2.9).

3.6.2 Terminals.

3.6.2.1 Terminal strength (see 4.6.3.1). Terminals shall reveal no evidence of breakage, loosening, or relative motion between the electrical terminals and the body of the unit.

3.6.2.2 Solderability (see 4.6.3.2). Terminal solderability shall be in accordance with MIL-STD-454, requirement 5.

3.6.3 System modularization and accessibility. The modular construction of the units shall provide for modular replacement at the organizational (shipboard) level. Modular construction of the units shall be used to facilitate the removal and replacement of components and assemblies while minimizing the removal of other system components or assemblies.

3.6.4 Standardization and interchangeability. The unit shall be constructed to minimize the quantity of special tools required for all levels of maintenance. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)). Standardized piece parts shall be used wherever possible. Circuit card assemblies shall be constructed to be interchangeable between the applicable unit (FDM, FDDM or MULDEM). The construction of the unit shall enable the unit to be functionally and physically interchangeable between different units of the same part number and shall be in accordance with MIL-STD-454, requirement 7.

3.6.5 Interconnect wire color (see 4.6.1). Interconnect wire color shall be in accordance with MIL-STD-104, or as specified (see 3.1).

3.6.6 Hardware. Unit hardware and unit mounting hardware shall be in accordance with MIL-E-16400 and shall be selected to support unit requirements for electromagnetic interference (EMI), electromagnetic compatibility, and durability.

3.6.6.1 Printed-wiring and printed-wiring assemblies.

3.6.6.1.1 Rigid-wiring boards. The construction of rigid printed-wiring shall be in accordance with MIL-STD-275.

3.6.6.1.2 Printed-wiring assemblies. The construction of printed wiring assemblies shall be in accordance with MIL-STD-1389, appendix F.

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3.6.6.1.3 Flexible printed-wiring assemblies. Flexible printed-wiring assemblies shall be as specified in MIL-P-50884, type I, class B.

3.6.6.2 Circuit-card edge connectors. Circuit-card edge connectors shall be as specified in MIL-C-28754 or MIL-C-28859.

3.6.6.3 Fiber optic transmitters. Fiber optic transmitters shall be as specified (see 3.1).

3.6.6.4 Fiber optic receivers. Fiber optic receivers shall be as specified (see 3.1).

3.6.6.5 Fiber optic pigtail. Units shall be supplied with a fiber or cable pigtail as specified (see 3.1). One end of the pigtail shall be permanently mounted in the unit case, and the mounting construction may provide strain relief for the pigtail in order to meet the mechanical requirements specified herein. The pigtail shall be not less than 1 meter. No metal components shall be used within the pigtail, except for mounting to the unit case or termination connector.

3.6.6.6 Fiber optic connectors. Fiber optic connectors shall be as specified (see 3.1).

3.6.7 Bonds and grounds for integral EMI suppressions. Units shall be designed and constructed with adequate provisions for bonding the unit to a subsystem ground plane. Bonding shall not be accomplished through screws connecting the equipment to mounting racks. Shock-mounted units or those employing vibration isolators shall use bonding straps to bypass the shock mount or isolator to achieve a low impedance bond, 2.5 milliohms or less, between the equipment and subsystem ground plane. Bonding jumpers shall be of the solid metal type and be as short as possible. However, in no case shall the length-to-width ratio of the jumper be in excess of 5 to 1. Bonding techniques employed shall not impede maintainability nor adversely affect interchangeability. Surfaces being bonded together shall be prepared by removing all anodic film, grease, paint, lacquer, dirt, or other foreign and high resistance materials or agents from the immediate area to ensure negligible radio frequency (RF) impedance between the adjacent metal parts. Upon completion of the bonding assembly and ascertainment of the specified 2.5 milliohms or less bonding impedance, the completed assembly shall be refinished in accordance with the original finish.

3.6.8 Human factors engineering. The units shall be constructed in accordance with ASTM F 1166, sections 33 through 40.

3.6.9 Selection of alternative materials, constructions, and parts. All of the materials, processes, or parts required for the fabrication of the units shall be selected in accordance with MIL-P-11268. Any item not in accordance with MIL-P-11268 shall be defined as a nonstandard part and shall be in accordance with nonstandard parts requirements as specified in MIL-STD-965. Procedures for obtaining approval for use of nonstandard items shall be in accordance with MIL-STD-454, requirement 22.

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3.6.10 Conductor identification (see 4.6.1). Conductor identification shall be in accordance with MIL-STD-454, requirement 20. Each non-insulated wire lead in excess of 10.2 centimeters (cm) shall be color-coded by means of colored lacquer located near terminals, except when the leads terminate at marked terminals (see 3.6.10.1) or when the terminal designation and the placement of the leads provide easy lead identification.

3.6.10.1 Terminal end identification. The ends of all conductors which terminate in lugs shall be clearly and permanently marked with the conductor identification. Markings shall be made on white synthetic resin tubing in accordance with MIL-I-631, type F, grade a, form U, class I, or on polyvinyl chloride pressure-sensitive adhesive marking tape. Wire markings shall be clearly visible in the assembled units. The tape markers shall be tightly wrapped with at least two turns around the wire.

3.6.11 Dimensions (see 4.6.1). The unit dimensions shall be as specified (see 3.1) and shall not exceed the limitations specified in MIL-E-16400.

3.6.12 Weight (see 4.6.1). The weight of the unit shall be as specified (see 3.1).

3.6.13 Finish (see 4.6.1). The unit housing finish shall not affect the legibility of the unit part number and required markings (see 3.16) over the life of the unit. The finish shall be in accordance with the applicable requirements specified in MIL-E-16400.

3.6.14 Redundancy. Unless otherwise specified (see 3.1), the units shall not require redundancy.

3.7 General performance (functional characteristics) (see 4.6.4). The units shall have functional characteristics as they pertain to the equipment function selected, as specified in 3.7.1 through 3.7.4. A typical configuration employing FDM's is shown on figure 1. Firmware specifications shall be in accordance with the requirements of DOD-STD-2167. There shall be no software in the unit.

3.7.1 Multiplexer. The FDM shall accept composite numbers and classes of band or rate limited electrical input signals (analog or digital) and provide a frequency-division multiplexed output signal (see 6.5.11) (electrical or optical) for transmission over a single wideband channel. The number of input and output channels shall be as specified (see 3.1). The FDM shall operate in conjunction with a complementary FDDM unit. The FDM shall include, but not be limited to, a subchannel interface element, a combiner/amplifier element, and a multiplexer interface element as shown on figure 2.

3.7.1.1 Subchannel input interface element. The subchannel input interface element shall provide the necessary signal conditioning and processing of the band or rate limited input signals. These functions shall include, but not be limited to, implementing the requisite modulation (see 6.5.21) formats and the appropriate isolation, equalization, and filtering of signals, as specified (see 3.1).

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3.7.1.2 Combiner/amplifier element. The combiner/amplifier element shall provide the requisite mixing of modulated signals and, as required, the frequency translation and processing of the composite frequency-division multiplexed signal, as specified (see 3.1).

3.7.1.3 Multiplexer output interface element. The multiplexer output interface element shall support the processing of the composite frequency-division multiplexed signal for transmission (electrical or optical) over a single wideband channel. Functions shall include, but not be limited to, the appropriate impedance matching (see 6.5.16) and signal isolation and conditioning, as specified (see 3.1). For multiplexers with an optical output signal, the multiplexer interface element shall also support the conversion of the electrical frequency-division multiplexed signal to a modulated optical signal.

3.7.2 Demultiplexer. The FDDM shall demultiplex frequency-division multiplexed signals (optical or electrical) and provide composite numbers and classes of band or rate limited electrical signals (see 6.5.9) (analog or digital). The number of input and output channels shall be as specified (see 3.1). The FDDM shall operate in conjunction with a complementary FDM unit. The FDDM shall include, but not be limited to, a subchannel interface element, a separator/amplifier element, and a demultiplexer interface element as shown on figure 3.

3.7.2.1 Subchannel output interface element. The subchannel output interface element shall support the requisite demodulation (see 6.5.7) of the frequency-division demultiplexed signal, as well as provide signal conditioning and equalization for the demodulated output signals, as specified (see 3.1).

3.7.2.2 Separator/amplifier element. The separator/amplifier element shall support the requisite separation of the frequency-division multiplexed signal, as specified (see 3.1).

3.7.2.3 Demultiplexer input interface element. The demultiplexer input interface element shall support interface requirements for the received frequency-division multiplexed signal as specified (see 3.1). For demultiplexers with an optical input signal, the demultiplexer interface element shall also support the recovery of the electrical frequency-division multiplexed signal from a modulated optical signal.

3.7.3 Multiplexer/demultiplexer (MULDEM). The MULDEM shall incorporate, within an integral unit, those functions inherent to the FDM and FDDM with an appropriate commonality of functions such as power distribution functions and supervisory signals (see 6.5.29). The MULDEM shall consist of a combination of those components for an FDM and FDDM as specified in 3.7.1 and 3.7.2 respectively, and 3.7.4.

3.7.4 Transmultiplexing. Transmultiplexing, from a time-division multiplexing (see 6.5.30) transmission medium, shall take place at an appropriate subchannel input port, as specified (see 3.1).

3.8 Detailed performance characteristics. Detailed operational and channel performance parameters and interface characteristics for multiplexing, demultiplexing, and transmultiplexing shall be as specified in 3.8.1 through 3.8.3.

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3.8.1 Operational performance parameters (see 4.6.5).

3.8.1.1 Prime power (see 4.6.5.1). The unit shall operate from prime power sources as specified (see 3.1).

3.8.1.1.1 Alternating current (ac) prime power variation (see 4.6.5.1.1). The unit shall operate within specified parameters defined in MIL-STD-1399, section 300.

3.8.1.1.2 Direct current (dc) prime power variation (see 4.6.5.1.2). The unit shall operate within specified maximum and minimum dc prime power voltage levels as specified (see 3.1).

3.8.1.1.3 Current load. The unit shall not exceed worst case current load requirements as specified (see 3.1).

3.8.1.1.4 Reverse polarity protection. The unit shall provide reverse polarity protection for dc power inputs. In such cases, the unit shall suffer no damage from, and shall provide specified performance following correction of the reverse polarity condition.

3.8.1.2 Alarm functions. The unit shall support the detection of fault conditions and provide requisite actions as consequence to these conditions as specified (see 3.1).

3.8.1.2.1 Fault conditions (see 4.6.5.2). The FDM unit shall detect the failure of prime power supply and loss of the incoming channels as specified (see 3.1). The FDDM unit shall detect the loss of the incoming multiplexed signal.

3.8.1.2.2 Consequent actions. Prompt maintenance alarms shall be provided to the fault conditions specified in 3.8.1.2.1.

3.8.1.2.2.1 Alarm indication signal (AIS) response time. When detecting the absence of a received channel, an AIS shall be inserted into the frequency slot as specified (see 3.1). The AIS shall be applied as soon as possible and not later than 2 milliseconds after detection of the fault condition.

3.8.1.3 Multiplexing/demultiplexing. The detailed operational performance parameters for implementing an FDM and FDDM shall be as specified in 3.8.1.3.1 through 3.8.1.3.10.

3.8.1.3.1 Subcarrier frequency (see 4.6.5.3.1). The subcarrier (see 6.5.3 and 6.5.28) frequency shall be sufficiently stable over environment and time that the required increase in pre-detection bandwidth due to frequency uncertainty shall be not more than 10 percent of the bandwidth possible with perfect frequency accuracy.

3.8.1.3.2 Harmonics (see 4.6.5.3.2 and 6.5.15). Unless otherwise specified (see 3.1), all harmonics of any subcarriers, as present within any unit channels

3.8 Detailed performance characteristics. Detailed operational and channel performance parameters and interface characteristics for multiplexing, demultiplexing, and transmultiplexing shall be as specified in 3.8.1 through 3.8.3.

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3.8.1.3.3 Spurious emissions (see 4.6.5.3.2 and 6.5.27). Unless otherwise specified (see 3.1), spurious emissions shall be not less than 55 dB below the fundamental frequency.

3.8.1.3.4 Carrier-to-noise ratio (see 4.6.5.3.3, 6.5.4, and 6.5.23). Unless otherwise specified (see 3.1), the carrier-to-noise ratio (CNR) shall be such that the noise level shall be not less than 50 dB below the unmodulated or continuous wave (CW) (see 6.4.6) subcarrier level within plus 5 percent of the unmodulated or CW subcarrier frequency.

3.8.1.3.5 Modulation and demodulation formats (see 4.6.5.3.4). The modulation and demodulation formats shall be as specified (see 3.1).

3.8.1.3.6 Guard-band width (see 4.6.5.3.5 and 6.5.13). Frequency guard bands shall be implemented in the FDM signal. Unless otherwise specified (see 3.1), the minimum guard-band width shall be equal to the sum of the maximum frequency drifts (see 6.5.12) of the adjacent subcarrier frequencies to which the guard band is located. Unless otherwise specified (see 3.1), frequency guard bands shall accommodate the necessary in-band signaling (see 6.5.17), such as pilot tones (see 6.5.25).

3.8.1.3.7 Propagation delay (see 4.6.5.3.6). The maximum signal delay from the input to the output of the unit shall be as specified (see 3.1).

3.8.1.3.8 Pilot tone characteristics (see 4.6.5.3.7).

3.8.1.3.8.1 Stability (see 4.6.5.3.7.1). Unless otherwise specified (see 3.1), the frequency stability (see 6.5.14) of the pilot tone shall remain within specified limits for at least 90 days without adjustment.

3.8.1.3.8.2 Level (see 4.6.5.3.7.2). Unless otherwise specified (see 3.1), the pilot tone level shall be minus 20 decibels referred to 1 milliwatt (dBmW) dBm with respect to a zero transmission level and with an accuracy of plus or minus 0.1 dB.

3.8.1.3.8.3 Level stability (see 4.6.5.3.7.3). The level stability of the pilot tone shall be plus or minus 0.3 dB for any 30-day period.

3.8.1.3.9 Frequency plan (see 4.6.5.3.8). The channel center frequencies and bandwidth shall be as specified (see 3.1). Low bandwidth signals shall be carried on low frequency channels and wide bandwidth channels shall be carried on high frequency channels. Channel spacings and frequencies shall be selected to reduce the need for low shape factor filters and minimize the effects of harmonic and intermodulation distortions, spurious emissions, and out-of-band noise.

3.8.1.3.10 Interchannel crosstalk (see 4.6.5.3.9). Interchannel crosstalk shall not exceed the level specified (see 3.1).

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3.8.1.4 Optical performance.

3.8.1.4.1 Dynamic range (see 4.6.5.4.1). The unit shall exhibit sufficient dynamic range to support operation under worst case conditions established for both minimum and maximum path length configurations, including connectors and repair splices, as specified (see 3.1), without degradation in performance and without operator interference.

3.8.2 Channel performance parameters (see 4.6.5.5).

3.8.2.1 Analog subchannels. The analog subchannel performance parameters for voice service, facsimile service, and quasi-analog signal (see 6.5.26) service shall be as specified (see 3.1). Performance requirements for additional types and classes of analog subchannel service shall be as specified (see 3.1).

3.8.2.2 Digital channels. Digital channel performance parameters shall be as specified (see 3.1).

3.8.2.3 Other signal types. Performance parameters for those signal types not addressed in this specification shall be as specified (see 3.1).

3.8.3 Interface characteristics (see 4.6.5.6).

3.8.3.1 Analog subchannel interface. Analog subchannel interface criteria for such parameters as impedance, signal levels, and equalization shall be as specified (see 3.1).

3.8.3.2 Digital channels. Digital signal interface parameters such as signal format, signal voltage levels, rise and fall times, clock and data relationship, short circuit protection, and data rate shall be as specified (see 3.1).

3.8.3.3 Multiplexed signal interface.

3.8.3.3.1 FDM signal output.

3.8.3.3.1.1 Electrical interface. The electrical signal interface characteristics shall be as specified (see 3.1). The electrical interface characteristics shall be compatible with performance parameters defined for the complementary FDDM electrical or optical input.

3.8.3.3.1.2 Optical interface. The optical signal interface parameters shall be as specified (see 3.1). The optical interface shall be compatible with performance parameters as defined for the complementary FDDM optical input.

3.8.3.3.2 FDDM signal input.

3.8.3.3.2.1 Electrical interface. The electrical signal interface characteristics shall be as specified (see 3.1). The electrical interface characteristics shall be compatible with performance parameters defined for the complementary FDM electrical or optical output.

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3.8.3.3.2.2 Optical interface. The optical signal interface parameters shall be as specified (see 3.1). The optical interface characteristics shall be compatible with performance parameters as defined for the complementary FDM optical output.

3.8.3.4 Maintenance orderwire. A maintenance orderwire shall be provided in accordance with requirements as specified (see 3.1).

3.8.3.5 Other interfaces. Interface criteria for those signals not addressed herein shall be as specified (see 3.1).

3.9 Environmental requirements. The units shall meet all requirements specified herein during the specified operating environments and after the specified storage environment. The operating temperature range and storage temperature range shall be as specified (see 3.1), as shown in table I.

TABLE I. Temperature ranges.

Temperature range	Operating (°C)	Non-operating (°C)
1	0 to +85	-62 to +85
2	-28 to +85	-62 to +85
3	-54 to +85	-62 to +85
4	0 to +125	-62 to +125
5	-28 to +125	-62 to +125
6	-54 to +125	-62 to +125

3.9.1 Component requirements (style A). Style A units shall meet the requirements specified in 3.9.1.1 through 3.9.1.9.

3.9.1.1 Temperature cycling (see 4.6.6.1.1). The unit shall meet the performance requirements over the operating temperature range. The operating temperature range shall be as specified (see 3.9) in accordance with table I. There shall be no evidence of physical damage.

3.9.1.2 Thermal shock (see 4.6.6.1.2). The unit shall be fully functional and show no evidence of physical degradation. The temperature extremes shall be the specified storage temperature extremes.

3.9.1.3 Moisture resistance (see 4.6.6.1.3). The unit shall meet the performance requirements specified herein.

3.9.1.4 Salt atmosphere (corrosion) (see 4.6.6.1.4). The unit shall meet the performance requirements specified herein.

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3.9.1.5 Mechanical shock (see 4.6.6.1.5). The unit shall meet the performance requirements specified herein.

3.9.1.6 Vibration (see 4.6.6.1.6). The unit shall meet the performance requirements specified herein.

3.9.1.7 Explosive atmosphere (see 4.6.6.1.7). The unit shall safely operate in flammable atmospheres without causing an explosion.

3.9.1.8 Nuclear radiation resistance (see 4.6.6.1.8). The unit shall meet the requirements as specified (see 3.1).

3.9.1.9 Environmental screening (see 4.3.1.3 and 4.3.1.4). The unit shall be environmentally screened by random vibration and by temperature cycling.

3.9.2 System/subsystem requirements (styles B and C). Styles B and C units shall meet the requirements specified in 3.9.2.1 through 3.9.2.13.

3.9.2.1 Temperature-humidity cycling (see 4.6.6.2.1). The unit shall sustain no damage or degradation in performance when subjected to a maximum relative humidity of 95 percent for both continuous and intermittent periods over the operating temperature range as specified (see 3.9), including conditions wherein condensation takes place, in and on the unit, in the form of both water and frost.

3.9.2.2 Non-operating temperature range (see 4.6.6.2.2). The unit shall sustain no damage or degradation in performance.

3.9.2.3 Salt spray (corrosion) (see 4.6.6.2.3). The unit shall be submitted to the effects of salt fog without sustaining any damage or degradation in performance. The base metal of the unit shall not be visible through the finish or coating, nor shall there be any evidence of blistering, softening, separation from the base metal, corrosion products, or other coating failure.

3.9.2.4 Fungus resistance (see 4.6.6.2.4). The unit shall meet the requirements of MIL-STD-454, requirement 4. Units that are not in accordance with MIL-STD-454, requirement 4 for fungus inert material, shall be in accordance with MIL-STD-810, method 508, grade 1 classification.

3.9.2.5 Icing (see 4.6.6.2.5). Racks and enclosures housing the unit shall sustain no structural damage when subjected to a maximum ice load of 21.9 kilograms per square meter (kg/m^2).

3.9.2.6 Shock (see 4.6.6.2.6). The unit shall have no cracks, breaks, or other physical damage.

3.9.2.7 Vibration (see 4.6.6.2.7). The unit shall have no cracks, breaks, or other physical damage.

3.9.2.8 Inclination (see 4.6.6.2.8). The unit shall operate satisfactorily at an inclination angle of 60 degrees.

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3.9.2.9 Water pressure (for watertight enclosures only) (see 4.6.6.2.9). The unit shall reveal no penetration of indicator dye.

3.9.2.10 Fluid immersion (see 4.6.6.2.10). The unit shall reveal no swelling or softening of material, no loss of sealing capabilities, or other effects detrimental to the operation of the unit.

3.9.2.11 Magnetic field environment (see 4.6.6.2.11). The unit shall be compatible with the magnetic field environment interface requirements specified in MIL-STD-1399, section 071.

3.9.2.12 Explosive atmosphere (see 4.6.6.2.12). The unit shall safely operate in flammable atmosphere without causing an explosion.

3.9.2.13 Nuclear radiation resistance (see 4.6.6.2.13). The unit shall meet the requirements specified (see 3.1).

3.10 Mechanical properties.

3.10.1 Compression resistance (for style B units only) (see 4.6.7.1). The unit shall not deform more than 1 percent from its largest cross-sectional dimension.

3.10.2 Impact resistance (for style B units only) (see 4.6.7.2). The unit shall reveal no physical damage.

3.10.3 Connector pull-out force (see 4.6.7.3). The minimum connector-to-unit pull-out force shall be one-half the specified tensile strength of the attached connector or 100 newtons, whichever is less. There shall be no evidence of connector-to-unit seal damage, distortion or bending of unit parts, or connector disengagement from the unit.

3.10.4 Cable pull-out force (see 4.6.7.4). The minimum cable-to-unit pull-out force shall be 50 percent of the specified tensile strength of the attached cable or 100 newtons, whichever is less. There shall be no evidence of cable jacket damage, cable clamp failure, cable-to-unit seal damage, distortion or bending of unit parts, or cable disengagement from the clamp.

3.10.5 Cable twist (see 4.6.7.5). Cable-to-unit seals shall not be rendered inoperable nor shall any other physical damage occur.

3.10.6 Cable seal flexing (see 4.6.7.6). Cable-to-unit strain relief mechanisms shall prevent loss of environmental sealing or other damage which may impair the unit operation.

3.10.7 Fiber dynamic strength (see 4.6.7.7). The minimum fiber dynamic strength shall be 50 percent of the specified proof strength of the attached fiber. There shall be no fiber-to-unit seal damage, or distortion or bending of unit parts.

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3.11 Electromagnetic interference suppression (see 4.6.8).

3.11.1 Electromagnetic. The unit shall meet the conducted and radiated emission requirements as specified (see 3.1), and the conducted and radiated susceptibility requirements of MIL-STD-461, part 5. Above deck or below deck shall be as specified (see 3.1).

3.11.1.1 Power supplies. Electromagnetic interference reduction in power supplies shall be in accordance with MIL-HDBK-241.

3.11.2 Lightning. Input suppression and transient suppression shall be incorporated in the unit as specified (see 3.1) to provide protection against lightning strikes and current and voltage transients.

3.11.3 Power leads. The unit shall sustain no damage or degradation in performance when subjected to voltage variations as specified (see 3.1).

3.12 Maintainability (for style B and C units only). Maintainability shall be in accordance with MIL-STD-454, requirement 54, and as specified herein. Maintainability shall use Built-In-Test-Equipment (BITE) and other features defined in 3.12.1 through 3.12.3 to detect, localize, and isolate failures.

3.12.1 Quantitative maintainability requirements. The maintainability requirement, expressed as the mean-time-to-repair (MTTR) (see 6.5.20), shall be as specified in 3.12.1.1 or as otherwise specified (see 3.1).

3.12.1.1 MTTR. The organizational level of maintenance shall have an MTTR of not greater than 20 minutes. The MTTR time figure includes localization (troubleshooting to modular level), isolation, disassembly, modular replacement, reassembly, and checkout of all corrective maintenance for restoration of system operation.

3.12.2 Maintainability prediction. Unless otherwise specified (see 3.1), a maintainability prediction shall be performed in accordance with MIL-HDBK-472, procedure II, part A.

3.12.3 Maintainability design criteria. Design features of styles B and C shall be included to:

(a) Minimize the number and frequency of required preventative maintenance actions.
(b) Provide adequate access to all components and minimize the requirements for special tools and test equipment. Ensure that each shipboard replaceable unit (SRU) shall be easily removable from its chassis or cabinet without the use of special tools.
(c) Ensure that it shall be possible to perform maintenance diagnostics with power applied, equipment open and operating for intervals not exceeding 15 minutes at ambient temperatures.
(d) Ensure that preventive maintenance at the organizational level be limited to visual equipment inspections, lubrications, mechanical parts, cleaning surfaces and interiors, and changing filters.
(e) Ensure that no intermediate level maintenance is required.
(f) Incorporate BITE.

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3.13 Maintenance. Style A units shall require no preventive or corrective maintenance. Styles B and C units shall require maintenance as specified herein.

3.13.1 Maintenance levels. Two levels of maintenance shall exist for the equipment described in this specification: organizational maintenance level and depot maintenance level.

3.13.1.1 Organizational maintenance level. The unit shall be designed to minimize the preventative and corrective maintenance workload at the organizational maintenance level. All required maintenance actions shall be compatible with the Third Class Interior Communications Technician (IC3) skill level or as specified (see 3.1).

3.13.1.1.1 Corrective maintenance. Corrective maintenance actions shall consist of SRU replacement at this level (see 3.12.1.1).

3.13.1.1.2 Preventive maintenance. Preventive maintenance at the organizational maintenance level shall be limited to visual equipment inspections, lubricating mechanical parts, cleaning surfaces and interiors, changing filters, and other similar tasks necessary to assure minimum equipment deterioration with age and use. Preventive maintenance shall not require the removal and replacement of components (except filters). Preventive maintenance requirements shall be grouped into periodic maintenance packages with the schedule interval maximized; the minimum allowable interval shall be greater than 90 days. Preventive maintenance shall be performed by one person. Whenever preventive maintenance is in progress, it shall be possible to restore the equipment to operational condition within 5 minutes. Requirements for preventive maintenance shall be such that a 1-month delay in performing a task will not adversely affect the operational performance of, or cause irrepairable damage to, the unit.

3.13.1.2 Depot maintenance level. The repair procedures, processes, materials, tools, and special purpose electronic test equipment required for corrective maintenance of the unit at the depot level, shall be determined by the Logistics Support Analysis (LSA). The units shall be designed to preclude the need for preventive maintenance at the depot maintenance level.

3.14 Safety. The handling, installation, and operation of the units shall not endanger personnel by direct contact or by long term effects due to deterioration of any component within the unit construction. The unit design and operation shall be in accordance with the safety standards of MIL-STD-454, requirement 1.

3.14.1 Terminals. Wire connections (bare terminals) shall use red polyolefin sleeving, as specified in MIL-I-23053 and MIL-I-23053/6, with nominal voltage and current type imprinted on sleeving.

3.14.2 Terminal strips. When specified (see 3.1), terminal strips shall be covered by red, transparent, nonconductive barriers in accordance with MIL-T-55164, with nominal voltage and current type identified. Small openings shall be provided where required for maintenance.

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3.14.3 Laser safety. Laser safety requirements shall be as specified (see 3.1).

3.14.4 Connector alignment. All connectors shall be provided with aligning pins or equivalent devices to aid in alignment, to preclude inserting in other than the desired position, and to preclude electrical shock.

3.15 Security. When specified (see 3.1), the unit shall provide full data security. The transmission security shall be in accordance with the TEMPEST requirements as specified (see 3.1).

3.16 Identification and marking (see 4.6.1).

3.16.1 Nomenclature (unit name and type designation). Unit name and type designation shall be established in accordance with the part numbering system specified in 6.8 and 6.9.

3.16.2 Serial numbers. Unless otherwise specified (see 3.1), serial numbers are required for each item of the equipment to which an identification plate is to be affixed. These serial numbers shall be in accordance with MIL-E-21981.

3.16.3 Marking. Marking of style A units shall meet the requirements of MIL-M-38510 and as specified herein. Marking of styles B and C units shall be in accordance with MIL-STD-454, requirement 67, and the applicable requirements as specified herein. Unless otherwise specified (see 3.1), marking shall conform to the normal service requirements of MIL-P-15024 and MIL-P-15024/5.

3.16.4 Identification plates; mounting and location. Identification plates and information plates shall be in accordance with MIL-P-15024 and MIL-P-15024/5 except that type G of MIL-P-15024 (adhesive-backed plates) shall not be used. Plates shall be mounted in a conspicuous space, generally on the front panel of the item to which it applies. When space does not permit mounting on the front or front panel, they may be mounted externally or internally in locations easily accessible. Labeling on units shall conform to MIL-STD-1472, except labels on units that incorporate a laser source shall be as specified (see 3.1). The Government-assigned identification and serial numbers shall be permanently marked on the unit.

3.16.4.1 Material compatibility. The material selected for identification plates and their fastening devices shall be compatible with the materials on which the plate is mounted.

3.17 Workmanship (see 4.6.1). Workmanship shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Screening (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Screening (for style A devices only). Each unit to be delivered or submitted for inspection shall have been subjected to and passed all of the screening tests in table II in the order shown. In order to properly perform the hermetic seal test, units with cable pigtails may be subjected to screening tests before the pigtail exterior is installed over the coated or buffered fiber. For those units with pigtails to be cabled after screening, a temporary strain relief may be installed at the unit-pigtail interface for the screening tests.

TABLE II. Screening inspection.

Inspection	Requirement	Test
Screening:		
Internal visual	3.5.1	4.3.1.1
Burn-in	3.3.2	4.3.1.2
Random vibration	3.9.1.9	4.3.1.3
Temperature cycling	3.9.1.9	4.3.1.4
External visual	3.5.1	4.3.1.5
Hermetic seal	3.5.1	4.3.1.6

4.3.1 Inspection conditions. All inspections shall be performed in accordance with the test conditions specified herein.

4.3.1.1 Internal visual (see 3.5.1). Internal visual inspection shall be performed in accordance with MIL-STD-883, method 2017.

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4.3.1.2 Burn-in (see 3.3.2). Burn-in testing shall be in accordance with MIL-STD-883, method 1015. Unless otherwise specified (see 3.1), all samples shall be tested for a minimum of 160 hours at the maximum temperature specified in 3.9. Maximum power supply voltage shall be applied and optical power level shall be at a maximum to or from the unit during test.

4.3.1.3 Random vibration (see 3.9.1.9). Random vibration testing shall be in accordance with MIL-STD-883, method 2026, condition II, test condition C. No power shall be applied during this test.

4.3.1.4 Temperature cycling (see 3.9.1.9). Temperature cycling shall be in accordance with MIL-STD-883, method 1010, using the non-operating temperature range (see 3.9). No power shall be applied during the test.

4.3.1.5 External visual (see 3.5.1). External visual inspection shall be in accordance with MIL-STD-883, method 2009.

4.3.1.6 Hermetic (see 3.5.1). The unit shall be tested in accordance with MIL-STD-883, method 1014. Test condition A1 shall be used for the seal test, and test condition C shall be used for the gross leak test.

4.4 First article inspection. First article inspection shall consist of all the tests listed in table III. The units submitted for testing shall be representative of the production process.

TABLE III. First article inspection.

Inspection	Requirement	Test	Sample size
Group I			
Identification and marking	3.16	4.6.1	4
Case	3.5.1	4.6.2.1	1/
Interconnect wire color	3.6.5	4.6.1	1/
Conductor identification	3.6.10	4.6.1	1/
Dimensions	3.6.11	4.6.1	1/
Weight	3.6.12	4.6.1	1/
Finish	3.5.3 3.6.13	4.6.1	1/
Functional characteristics	3.7	4.6.4	1/

See footnotes at end of table.

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TABLE III. First article inspection - Continued.

Inspection	Requirement	Test	Sample size
Operational performance	3.8.1	4.6.5	1/
Channel performance	3.8.2	4.6.5.5	1/
Interface characteristics	3.8.3	4.6.5.6	1/
Workmanship	3.17	4.6.1	1/
Group II			
Terminal strength	3.5.2.1 3.6.2.1	4.6.2.2 4.6.3.1	1
Solderability	3.5.2.2 3.6.2.2	4.6.2.3 4.6.3.2	2/
Temperature cycling	3.9.1.1	4.6.6.1.1	1
Thermal shock	3.9.1.2	4.6.6.1.2	3/
Moisture resistance	3.9.1.3	4.6.6.1.3	3/
Salt atmosphere	3.9.1.4	4.6.6.1.4	3/
Mechanical shock	3.9.1.5	4.6.6.1.5	3/
Vibration	3.9.1.6	4.6.6.1.6	3/
Temperature humidity cycling	3.9.2.1	4.6.6.2.1	3/
Non-operating temperature range	3.9.2.2	4.6.6.2.2	3/
Salt spray	3.9.2.3	4.6.6.2.3	3/
Fungus resistance	3.9.2.4	4.6.6.2.4	3/
Icing	3.9.2.5	4.6.6.2.5	3/
Shock	3.9.2.6	4.6.6.2.6	3/
Vibration	3.9.2.7	4.6.6.2.7	3/
Inclination	3.9.2.8	4.6.6.2.8	3/

See footnotes at end of table.

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TABLE III. First article inspection - Continued.

Inspection	Requirement	Test	Sample size
Water pressure	3.9.2.9	4.6.6.2.9	3/
Fluid immersion	3.9.2.10	4.6.6.2.10	3/
Group III			
Compression resistance	3.10.1	4.6.7.1	1
Impact resistance	3.10.2	4.6.7.2	4/
Connector pull-out force	3.10.3	4.6.7.3	4/
Cable pull-out force	3.10.4	4.6.7.4	4/
Cable twist	3.10.5	4.6.7.5	4/
Cable seal flexing	3.10.6	4.6.7.6	4/
Fiber dynamic strength	3.10.7	4.6.7.7	4/
Group IV			
Electromagnetic interference suppression	3.11	4.6.8	1
ESDS	3.5.4	4.6.2.4	5/
Explosive atmosphere	3.9.2.12	4.6.6.2.12	5/
Magnetic field environment	3.9.2.11	4.6.6.2.11	5/
Nuclear radiation resistance	3.9.1.8 3.9.2.13	4.6.6.1.8 4.6.6.2.13	5/

1/ The same samples shall be used as in the identification and marking inspection.
2/ The same sample shall be used as in the terminal strength inspection.
3/ The same sample shall be used as in the temperature cycling inspection.
4/ The same sample shall be used as in the compression resistance inspection.
5/ The same sample shall be used as in the electromagnetic interference inspection.

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4.4.1 Sample. A unit shall be submitted for each unit construction for which first article approval is desired. The sample submitted shall be four units of the same part number.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table III in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided for testing in the remaining three groups. After completion of sample testing, all units shall be resubjected to group I testing. Any unit failing any inspection shall not be subjected to further inspection.

4.4.3 Failures. One or more failures shall be sufficient cause for refusal to grant first article approval. When first article test units are taken from a larger lot, and the first article is disapproved, none of the units from that lot shall be delivered.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections and tests specified for group A inspection (table IV), group B inspection (table V), and group C inspection (table VI), as specified (see 6.4).

4.5.1 Group A inspection. Group A inspections shall follow the order shown in table IV.

TABLE IV. Group A inspection.

Inspection	Requirement	Test
Identification and marking	3.16	4.6.1
Case	3.5.1	4.6.2.1
Interconnect wire color	3.6.5	4.6.1
Conductor identification	3.6.10	4.6.1
Dimensions	3.6.11	4.6.1
Weight	3.6.12	4.6.1
Finish	3.5.3 3.6.13	4.6.1

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TABLE IV. Group A inspection - Continued.

Inspection	Requirement	Test
Functional characteristics	3.7	4.6.4
Operational performance	3.8.1	4.6.5
Channel performance	3.8.2	4.6.5.5
Interface characteristics	3.8.3	4.6.5.6
Workmanship	3.17	4.6.1

4.5.2 Group B inspection. Group B inspection shall consist of the inspections specified in table V. Group B inspections shall be made on units that have passed the group A inspection.

TABLE V. Group B inspection.

Inspection	Requirement	Test
Non-operating temperature range	3.9.2.2	4.6.6.2.2
Shock	3.9.2.6	4.6.6.2.6
Inclination	3.9.2.8	4.6.6.2.8
Thermal shock	3.9.1.2	4.6.6.1.2
Mechanical shock	3.9.1.5	4.6.6.1.5

4.5.3 Group C inspections. Group C inspections shall consist of the inspections in table VI. Group C samples shall have satisfactorily completed all group A and group B inspections. After completion of group C inspection, the samples shall be re-subjected to group A inspection.

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TABLE VI. Group C inspection.

Inspection	Requirement	Test
Terminal strength	3.5.2.1 3.6.2.1	4.6.2.2 4.6.3.1
Solderability	3.5.2.2 3.6.2.2	4.6.2.3 4.6.3.2
Temperature cycling	3.9.1.1	4.6.6.1.1
Moisture resistance	3.9.1.3	4.6.6.1.3
Salt atmosphere	3.9.1.4	4.6.6.1.4
Vibration	3.9.1.6	4.6.6.1.6
Temperature humidity cycling	3.9.2.1	4.6.6.2.1
Salt spray	3.9.2.3	4.6.6.2.3
Fungus resistance	3.9.2.4	4.6.6.2.4
Icing	3.9.2.5	4.6.6.2.5
Vibration	3.9.2.7	4.6.6.2.7
Water pressure	3.9.2.9	4.6.6.2.9
Fluid immersion	3.9.2.10	4.6.6.2.10
Compression resistance	3.10.1	4.6.7.1
Impact resistance	3.10.2	4.6.7.2
Connector pull-out force	3.10.3	4.6.7.3
Cable pull-out force	3.10.4	4.6.7.4
Cable twist	3.10.5	4.6.7.5
Cable seal flexing	3.10.6	4.6.7.6
Fiber dynamic strength	3.10.7	4.6.7.7

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TABLE VI. Group C inspection - Continued.

Inspection	Requirement	Test
Electromagnetic interference suppression	3.11	4.6.8
ESDS	3.5.4	4.6.2.4
Explosive atmosphere	3.9.2.12	4.6.6.2.12
Magnetic field environment	3.9.2.11	4.6.6.2.11
Nuclear radiation resistance	3.9.1.8 3.9.2.13	4.6.6.1.8 4.6.6.2.13

4.5.3.1 Disposition of sample units. Sample units that have failed shall not be delivered.

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection. Units shall be examined to ascertain that the material construction (assembly, dimensions, weight, color, and finish), markings, and workmanship conform to the applicable requirements. Examinations shall be limited to those that may be performed without disassembling the unit in such a manner that its performance, durability, or appearance will be affected.

4.6.2 Design and construction inspection (style A units only).

4.6.2.1 Visual (see 3.5.1). The units' shall be visually inspected in accordance with MIL-STD-883, method 2009.

4.6.2.2 Terminal strength (see 3.5.2.1). The units electrical terminals shall be tested in accordance with MIL-STD-883, method 2004. Test condition A shall be used for tension; condition B1 shall be used for stress; and condition B2 shall be used for lead fatigue.

4.6.2.3 Solderability (see 3.5.2.2). Testing shall be performed on the units' electrical leads in accordance with MIL-STD-883, method 2003 (see 3.6.2.2).

4.6.2.4 Electrostatic discharge sensitivity (see 3.5.4). The unit shall be tested for sensitivity to electrostatic discharge in accordance with MIL-M-38510.

4.6.3 Design and construction inspection (styles B and C units only).

4.6.3.1 Terminal strength (see 3.6.2.1). The terminals shall be tested in accordance with MIL-STD-202, method 211. Test condition A and test condition E shall be used. The load for test condition A shall be 4.5 kilograms (kg).

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4.6.3.2 Solderability (see 3.6.2.2). The terminals shall be tested in accordance with MIL-STD-202, method 208.

4.6.4 Functional characteristics tests. Functional characteristics of the unit shall be demonstrated to verify the requirements specified in 3.7.

4.6.5 Operational tests (see 3.8.1).

4.6.5.1 Prime power tests.

4.6.5.1.1 AC prime power variation (see 3.8.1.1.1). Units shall be supplied with ac prime power in accordance with MIL-STD-1399, section 300 and operation verified.

4.6.5.1.2 DC prime power variation. Units shall be supplied nominal maximum and minimum dc prime power as specified (see 3.8.1.1.2), and operation verified.

4.6.5.2 Fault conditions (see 3.8.1.2.1). The fault conditions test shall include the removal of prime power and the composite multiplexed signal with the unit response verified.

4.6.5.3 Multiplexing/demultiplexing.

4.6.5.3.1 Subcarrier frequency (see 3.8.1.3.1). For a carrier in the frequency range of 0.014 MHz to 12 GHz, carrier stability for nonpulsed or pulsed modulation shall be verified in accordance with MIL-STD-449, method CE105, and MIL-STD-449, method CE106, respectively. Other carrier frequencies shall be tested if specified (see 3.1).

4.6.5.3.2 Emission spectrum characteristics (see 3.8.1.3.2 and 3.8.1.3.3). For an operating frequency range of 0.014 MHz to 12 GHz, requirements for harmonics and spurious emissions, narrowband and wideband (nonpulsed or pulsed modulation), shall be verified in accordance with the following:

(a) Nonpulsed, narrowband, MIL-STD-449, method CE107.
(b) Pulsed, narrowband, MIL-STD-449, method CE108.
(c) Nonpulsed, wideband, MIL-STD-449, method CE109.
(d) Pulsed, wideband, MIL-STD-449, method CE110.

At other operating frequencies, tests shall be as specified (see 3.1).

4.6.5.3.3 Carrier-to-noise ratio (see 3.8.1.3.4). All FDM channels, except the channel under test, shall have information modulated onto their respective carriers such that the full channel bandwidth is occupied. The carrier power level and the noise power level shall be measured with a spectrum analyzer on the channel under test. The CNR for the channel under test shall be calculated as follows:

$$CNR = 10\log_{10} (\text{Carrier power} / \text{Noise power})$$

The procedure shall be repeated for all remaining channels.

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4.6.5.3.4 Modulation and demodulation formats (see 3.8.1.3.5). The modulation and demodulation formats shall be verified over the specified frequency range in accordance with MIL-STD-449, method CE102.

4.6.5.3.5 Guard-band width (see 3.8.1.3.6). The guard-band width shall be calculated as the difference between the channel bandwidth and the frequency spacing of adjacent channels. The bandwidth and frequency spacing shall be measured in accordance with MIL-STD-449, method CE109.

4.6.5.3.6 Propagation delay (see 3.8.1.3.7). Propagation delay shall be verified in accordance with EIA 455-118. Delay through the test equipment and test cable shall be taken into account.

4.6.5.3.7 Pilot tones (see 3.8.1.3.8).

4.6.5.3.7.1 Stability (see 3.8.1.3.8.1). For a pilot tone in the frequency range of 0.014 MHz to 12 GHz, pilot tone stability for nonpulsed or pulsed modulation shall be verified in accordance with MIL-STD-449, method CE105, and MIL-STD-449, method CE106, respectively. Other pilot tone frequencies shall be tested as specified (see 3.1).

4.6.5.3.7.2 Level (see 3.8.1.3.8.2). The pilot tone level shall be measured as specified (see 3.1).

4.6.5.3.7.3 Level stability (see 3.8.1.3.8.3). The level stability of the pilot tone shall be measured as specified (see 3.1).

4.6.5.3.8 Frequency plan (see 3.8.1.3.9). The frequency spacing of each channel shall be verified over the entire frequency range of the unit in accordance with MIL-STD-449, method CE109.

4.6.5.3.9 Interchannel crosstalk (see 3.8.1.3.10). The input to the FDDM shall have carriers present on all channels, with only one channel selected to carry information. The power level shall be measured on the output of each channel. The interchannel crosstalk of two channels shall be calculated as follows:

$$\text{Crosstalk (dB)} = 10\log_{10} \frac{\text{(power level without information)}}{\text{(power level with information)}}$$

The procedure shall be repeated for all remaining channels.

4.6.5.4 Optical performance.

4.6.5.4.1 Dynamic range. Dynamic range shall be verified by proper operation at the specified maximum and minimum optical path configurations as specified (see 3.8.1.4.1).

4.6.5.5 Channel performance parameters (see 3.8.2). Channel performance parameters for the multiplexer shall be measured in accordance with MIL-STD-449, methods CE101, CE102, CE103, CE105, CE107, CE109, and CE111, as applicable. Channel performance parameters for the demultiplexer shall be measured in accordance with MIL-STD-449, methods CS101, CS103, CS106, CS108, CS110, CS112, CS114, and CS116, as applicable.

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4.6.5.6 Interface characteristics (see 3.8.3). Interface characteristics, such as power levels, shall be measured in accordance with MIL-STD-449, methods CE103 and CS103, as applicable.

4.6.6 Environmental tests. For tests in which units are operational, a unit fault during an environmental test shall be considered a failure.

4.6.6.1 Style A test methods.

4.6.6.1.1 Temperature cycling (see 3.9.1.1). The unit shall be tested in accordance with MIL-STD-883, method 1010.

4.6.6.1.2 Thermal shock (see 3.9.1.2). The unit shall be tested in accordance with MIL-STD-883, method 1011.

4.6.6.1.3 Moisture resistance (see 3.9.1.3). The unit shall be tested in accordance with MIL-STD-883, method 1004.

4.6.6.1.4 Salt atmosphere (corrosion) (see 3.9.1.4). The unit shall be tested in accordance with MIL-STD-883, method 1009.

4.6.6.1.5 Mechanical shock (see 3.9.1.5). The unit shall be tested in accordance with MIL-STD-883, method 2002, test condition B.

4.6.6.1.6 Vibration (see 3.9.1.6). The unit shall be tested in accordance with MIL-STD-883, test condition II and test condition C, method 2006; and test condition A, method 2007.

4.6.6.1.7 Explosive atmosphere (see 3.9.1.7). The unit shall be tested in accordance with MIL-STD-202, method 109, under conditions specified (see 3.1).

4.6.6.1.8 Nuclear radiation resistance. The unit shall be subjected to radiation in accordance with MIL-STD-883, methods 1017 (neutron) and 1019 (total dose). The dose rates and exposure times shall be as specified (see 6.2). Three test temperatures are required; the low and high operating temperatures (see 3.9), and 20 degrees Celsius (°C). The unit shall meet the performance requirements specified (see 3.9.1.8).

4.6.6.2 Styles B and C test methods.

4.6.6.2.1 Temperature-humidity cycling (see 3.9.2.1). The unit shall be tested in accordance with MIL-STD-810, method 507, procedure III.

4.6.6.2.2 Non-operating temperature range (see 3.9.2.2). The unit shall be tested in accordance with MIL-STD-810, method 501, procedure I and MIL-STD-810, method 502, procedure I, using the temperature ranges specified (see 3.9) in accordance with table I.

4.6.6.2.3 Salt spray (see 3.9.2.3). The unit shall be tested in accordance with MIL-STD-810, method 509, procedure I, for 96 hours.

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4.6.6.2.4 Fungus resistance (see 3.9.2.4). Units that do not meet the requirements of fungus-inert materials in accordance with MIL-STD-454, requirement 4, shall be tested by exposure to fungus in accordance with MIL-STD-810, method 508.

4.6.6.2.5 Icing. The unit shall be tested by demonstration of operation under ice loads as specified (see 3.9.2.5).

4.6.6.2.6 Shock (see 3.9.2.6). Shock requirements shall be verified in accordance with MIL-S-901, grade A, type A, class I.

4.6.6.2.7 Vibration (see 3.9.2.7). Vibration requirements shall be verified in accordance with type I vibration test of MIL-STD-167-1.

4.6.6.2.8 Inclination. The unit shall be tested by demonstration of operation under various inclination angles up to and including the requirements specified (see 3.9.2.8).

4.6.6.2.9 Water pressure (see 3.9.2.9). The unit shall be immersed in an aqueous dye penetrant solution at an applied pressure of 0.1 megapascal (equivalent to a depth of 10.4 meters) for 48 hours at a temperature of 10 to 35°C.

4.6.6.2.10 Fluid immersion (see 3.9.2.10). The unit shall be tested in accordance with EIA 455-12. Each unit shall be immersed in each immersion fluid as specified in table VII for a period of 24 hours. Each unit shall be completely dried before further testing.

TABLE VII. Immersion test fluids.

Fluids	Specification	Test temperature (°C)
Fuel oil	MIL-F-16884	33 - 37
Turbine fuel (JP-5)	MIL-T-5624	20 - 25
Isopropyl alcohol	TT-I-735	20 - 25
Hydraulic fluids	MIL-H-5606	48 - 50
Lubricating oils	MIL-L-17331 MIL-L-23699	73 - 77
Coolant 1/	-	20 - 25
Seawater	-	20 - 25

1/ Monsanto Coolanol 25 or equivalent.

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4.6.6.2.11 Magnetic field environment (see 3.9.2.11). The unit shall be tested in accordance with MIL-STD-462 as specified (see 3.1).

4.6.6.2.12 Explosive atmosphere (see 3.9.2.12). Safe operation of the unit within explosive atmospheric environments shall be tested using MIL-STD-202, method 109, under conditions as specified (see 3.1).

4.6.6.2.13 Nuclear radiation resistance (see 3.9.2.13). Units shall be tested for effects of nuclear radiation as specified (see 3.1). The dose rates and exposure times shall be as specified (see 6.2).

4.6.7 Mechanical.

4.6.7.1 Compression resistance (for style B units only) (see 3.10.1). The unit shall be tested as follows: A force of 890 newtons shall be applied uniformly over each of its sides and shall be maintained for 10 minutes. The test shall be performed at the maximum and minimum operating temperatures.

4.6.7.2 Impact resistance (for style B units only) (see 3.10.2). The unit shall be fully operating during and after an impact test of 13.6 newtons at the specified minimum and maximum operating temperatures (see 3.9).

4.6.7.3 Connector pull-out force (see 3.10.3). The unit shall be tested in accordance with EIA 455-6, with the exception that the test sample shall be a connector on a unit. The axial tensile load shall be applied and maintained for 10 minutes.

4.6.7.4 Cable pull-out force (see 3.10.4). The unit shall be tested in accordance with EIA 455-6. The axial tensile load shall be applied and maintained for 10 minutes.

4.6.7.5 Cable twist (see 3.10.5). The unit shall be tested in accordance with EIA 455-36, with the exception that the test sample shall be a properly pigtailed unit. The tensile load shall be 5 newtons and the number of loads to be applied shall be one.

4.6.7.6 Cable seal flexing (see 3.10.6). The unit shall be tested in accordance with MIL-STD-1344, method 2017, with the exception that the test sample shall be a properly pigtailed unit.

4.6.7.7 Fiber dynamic strength (see 3.10.7). The fiber pigtail dynamic strength shall be tested as follows: The fiber pigtail shall have an axial tensile load applied up to the load specified at an angle of 45 degrees to the normal (see figure 4). With the load applied, the pigtail shall be rotated through one rotation (360 degrees).

4.6.8 Electromagnetic interference suppression (see 3.11). The unit shall be tested in accordance with MIL-STD-462 as specified (see 3.1).

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packaging. The preservation, packing, and marking shall be in accordance with MIL-E-17555 (see 6.2).

5.1.1 Marking.

5.1.1.1 Standard marking. In addition to any special or other identification marking required, each unit supplementary, intermediate and exterior container shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or part number, as applicable (including the CAGE), shall be marked on all units, supplementary and intermediate packs in accordance with the identification marking provisions specified in MIL-STD-129.

5.1.1.2 Special marking. All units, supplementary, intermediate and exterior containers shall be marked as specified for sensitive electronic devices in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The units, in conjunction with fiber optic or metallic cable assemblies, are intended to be utilized to communicate information between remote areas aboard Naval ships.

6.1.1 FDM. FDM units, along with analog fiber optic transmitters, are intended to be employed to provide the appropriate channel interface, multiplex the electrical signals, perform electrical to optical signal conversion, and support the transmission of an optically modulated signal. In the electrical realm, intended FDM port communication includes higher-order multiplexers, either FDM or TDM (transmultiplexing) and fiber optic transmitters as shown on figure 1.

6.1.2 FDDM. FDDM units, along with analog fiber optic receivers, are intended to be employed to support the reception of an optically modulated signal, optical to electrical signal conversion, the demultiplexing of the signal into its electrical digital and analog channels, and routing to the appropriate channel interface. In the electrical realm, intended FDDM port communication includes higher-order FDDM's, either FDDM or TDDM (transmultiplexing) and fiber optic receivers as inferred from the complement of figure 1.

6.1.3 MULDEM. MULDEM's are integral units functionally comprised of a FDM and FDDM that are in duplex operation at a single node of a transmission system.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Applicable specification sheet number, title, and date.
(d) When first article is required (see 3.2).
(e) Nuclear radiation dose rates and exposure times, when test is required (see 4.6.6.1.8 and 4.6.6.2.13).
(f) Level of packaging required (see 5.1).
(g) Special marking if required (see 5.1.1.1).
(h) Applicable part number (see 6.8).
(i) Quantity of units required.

6.3 First article. When first article inspection is required, the items should be a first article sample. The first article should consist of 4 units. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.5 Definitions.

6.5.1 Analog signal. An analog signal is a nominally continuous electrical signal that varies in a direct correlation to the instantaneous value of a physical variable.

6.5.2 Bandwidth. The bandwidth is the difference between the limiting frequencies within which performance of a device, in respect to some characteristic, falls.

6.5.3 Carrier. A carrier is a wave, pulse train, or other signal suitable for modulation by an information-bearing signal in a modulator for the purpose of representing information and capable of being transported or transmitted by a communication system.

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6.5.4 Carrier-to-noise ratio (CNR). The CNR is the ratio, in decibels, of the value of the carrier to that of the noise in the receiver intermediate frequency bandwidth before any nonlinear process such as amplitude limiting and detection.

6.5.5 Channel. A channel is a path along which signals can be sent.

6.5.6 Continuous wave (CW). A continuous wave is a radio wave of constant amplitude and constant frequency.

6.5.7 Demodulation. Demodulation is the process wherein a signal resulting from previous modulation is processed to derive a wave having substantially the characteristics of the original modulating signal.

6.5.8 Frequency-division demultiplexer (FDDM). A unit responsible for performing the frequency-division demultiplexing process.

6.5.9 Frequency-division demultiplexing. Frequency-division demultiplexing is the complement of frequency-division multiplexing.

6.5.10 Frequency-division multiplexer (FDM). A unit responsible for performing the frequency-division multiplexing process.

6.5.11 Frequency-division multiplexing. Frequency-division multiplexing is a multiplexing scheme in which the available transmission frequency range is divided into narrow frequency bands, each used as a separate channel.

6.5.12 Frequency drift. Frequency drift is a slow, undesired change in the frequency of an oscillator, transmitter, receiver, or other equipment.

6.5.13 Frequency guard band. A frequency guard band is a frequency band left vacant between two channels to provide a margin of safety against mutual interference.

6.5.14 Frequency stability. Frequency stability is a measure of the variations of the frequency of an oscillator from its mean frequency over a specified period of time.

6.5.15 Harmonic. Harmonic is the presence of frequencies at the output of a device, which are harmonically related to a single frequency applied to the input of the device.

6.5.16 Impedance matching. Impedance matching is the connection of an additional impedance to an existing one in order to improve performance or to accomplish a specific effect.

6.5.17 In-band signaling. In-band signaling is signaling which uses frequencies or time slots within the bandwidth of the information channel.

6.5.18 Interface. An interface is a concept involving the interconnection between two devices of systems.

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6.5.19 Mean-time-between-failures (MTBF). MTBF for a particular measurement interval is the total functional life of a population of an item divided by the total number of failures within the population during the measurement interval.

6.5.20 Mean-time-to-repair (MTTR). MTTR is the total corrective maintenance time, that is, the total time devoted to maintenance, divided by the total number of maintenance actions, during a given time period.

6.5.21 Modulation. Modulation is the process, or result of the process, of varying certain characteristics of a signal, called a carrier, in accordance with a message signal.

6.5.22 Multiplex. Multiplex refers to the use of a common channel to make two or more channels, either by splitting of the frequency band transmitted by the common channel into narrower bands, each of which is used to constitute a distinct channel (frequency-division multiplex), or by allotting this common channel to multiple users in turn, to constitute different intermittent channels (time-division multiplex).

6.5.23 Noise. Noise is an undesired disturbance within the useful frequency band; the summation of unwanted or disturbing energy introduced into a communications system from man-made and natural sources.

6.5.24 Nonreturn-to-zero (NRZ) code. NRZ code is a code form having two states termed zero and one, and no neutral or rest condition.

6.5.25 Pilot tone. The pilot tone is a signal, usually a single frequency, transmitted over a system for supervisory, control, synchronization, or reference purposes.

6.5.26 Quasi-analog signal. A quasi-analog signal is a digital signal which has been converted to a form suitable for transmission over a specified analog channel.

6.5.27 Spurious emission. Spurious emission is any electromagnetic emission on a frequency or frequencies which are outside the necessary emission bandwidth, the level of which may be reduced without affecting the corresponding transmission of information.

6.5.28 Subcarrier. A subcarrier is a carrier which is applied as modulation on another carrier, or on an intermediate subcarrier.

6.5.29 Supervisory signal. A supervisory signal is a signal used to indicate (or in modern usage, to indicate and to control) the various operating states of the circuits or circuit combinations involved in a particular connection.

6.5.30 Time-division multiplexing. Time-division multiplexing is a method of deriving two or more apparently simultaneous channels from a given frequency spectrum of a propagation medium connecting two or more points by assigning discrete time intervals in sequence to each of the individual channels.

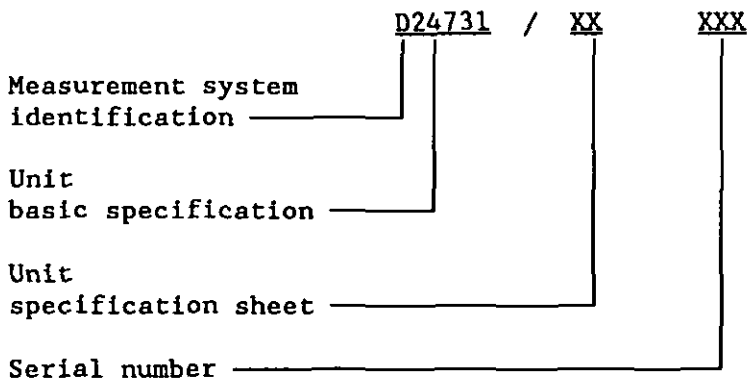
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6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Part or Identifying Number (PIN). Units should contain only the following:



Examples:

D24731/01-001
D24731/01-002

6.9 Part designator. A part designator, if specified (see 3.1), should include classifications (see 1.2) as follows:

(a) Equipment function (see 1.2.1).
(b) Multiplexed signal type (see 1.2.2).
(c) Channel configuration (see 1.2.3).
(d) Analog subchannel class (see 1.2.4).
(e) Digital subchannel class (see 1.2.5).
(f) Style (see 1.2.6).

Example: CCAIJB.

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6.10 Subject term (key word) listing.

Analog signal
Carrier
Channel
Digital signal
FDM
Fiber optic communications
Fiber optic component
Interface
Modulation
Optical signal
Signal
Subcarrier

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6008-N002)

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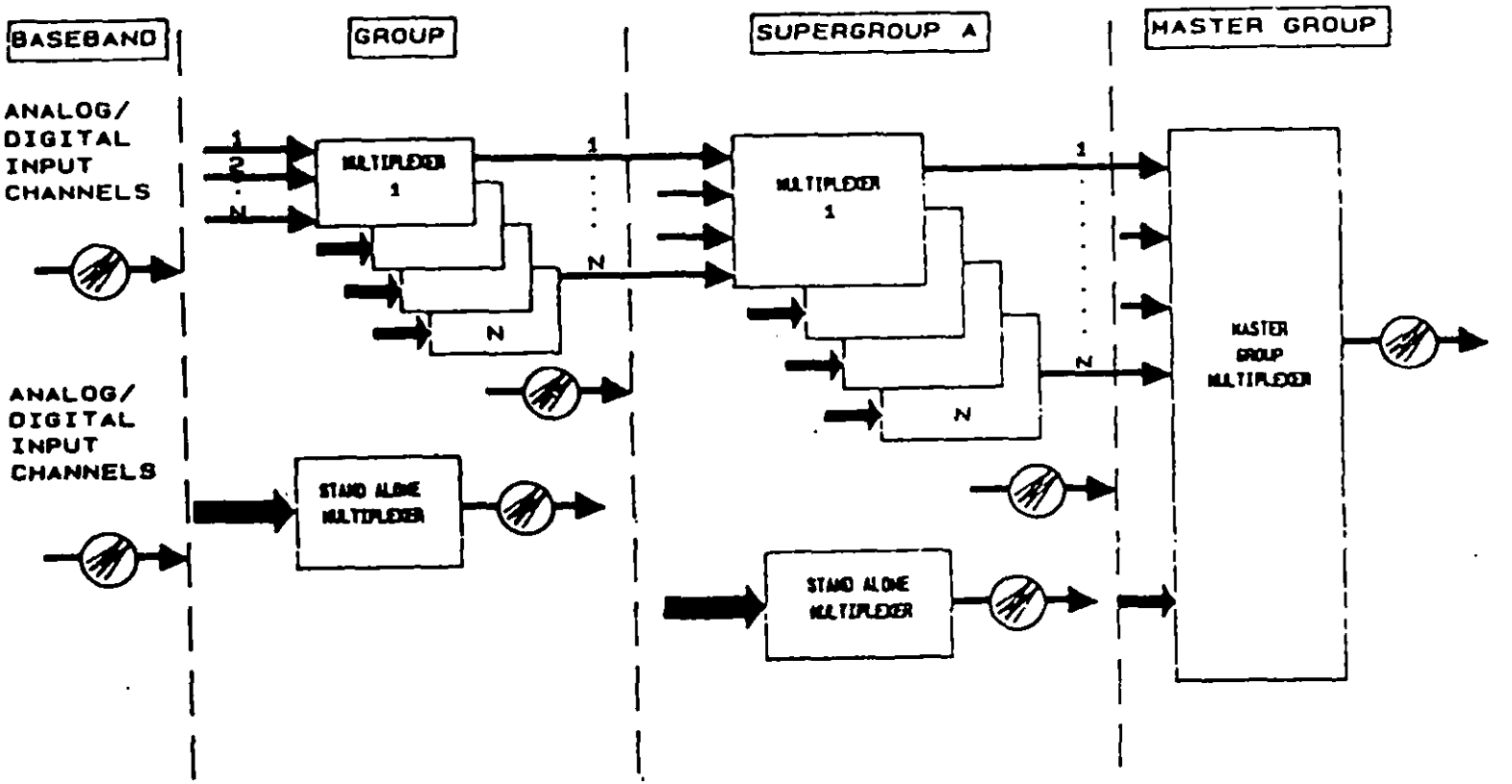


FIGURE 1. Typical multiplexer configuration.

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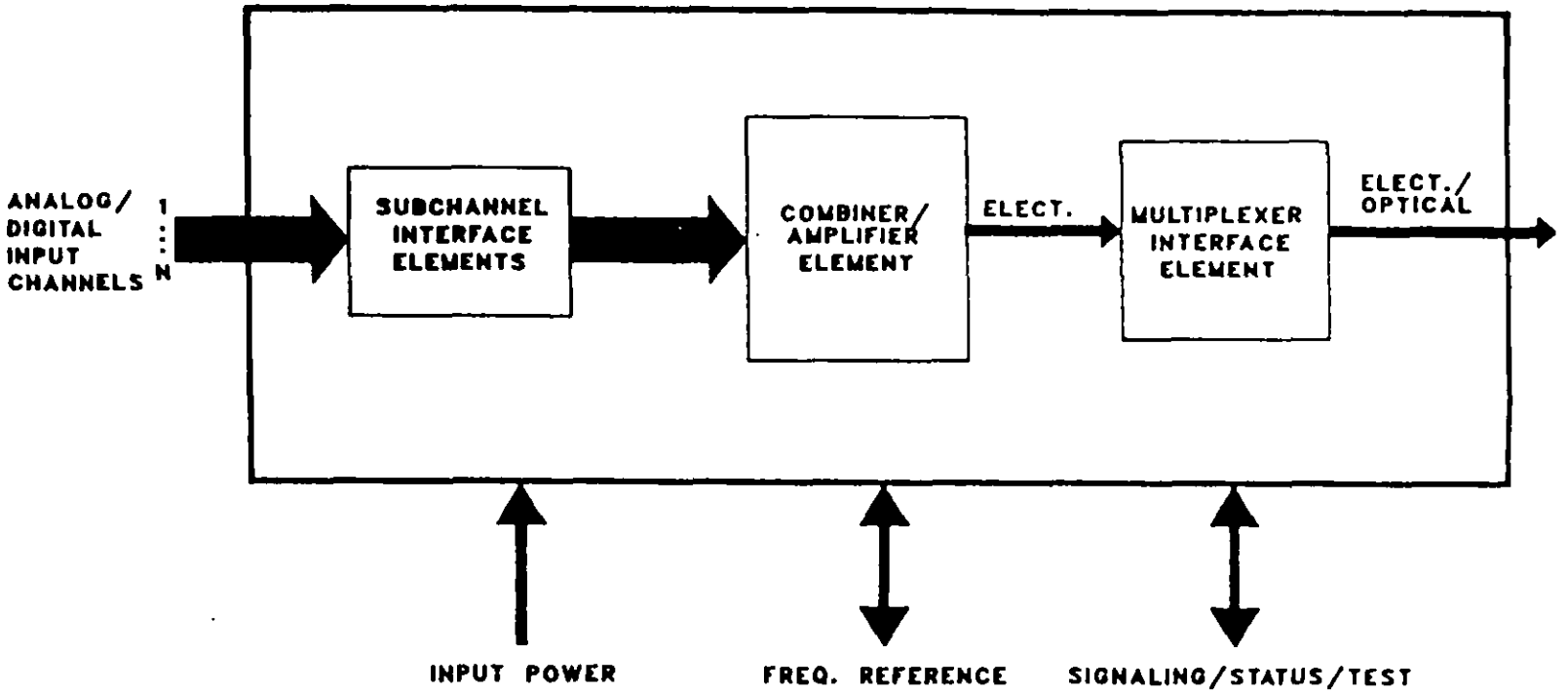


FIGURE 2. FDM conceptual block diagram.

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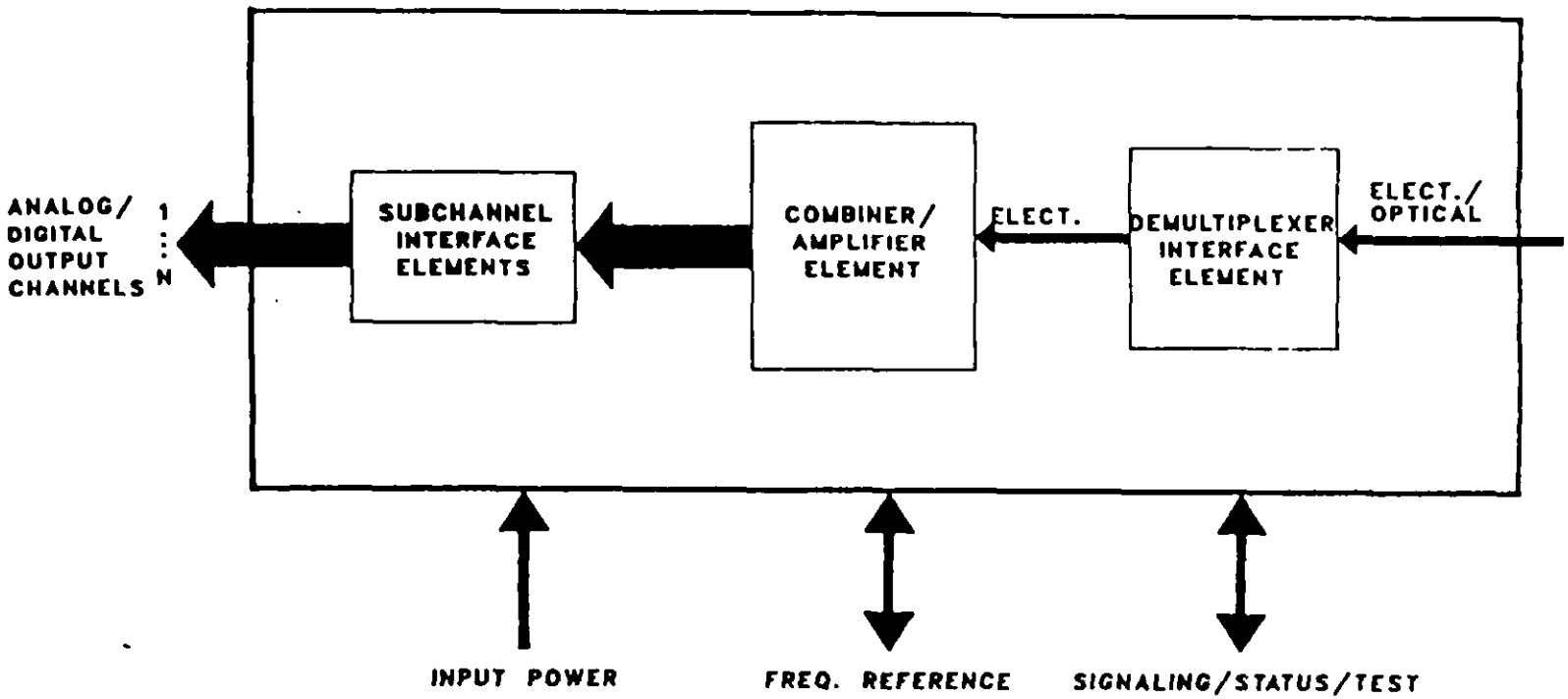


FIGURE 3. FDDM conceptual block diagram.

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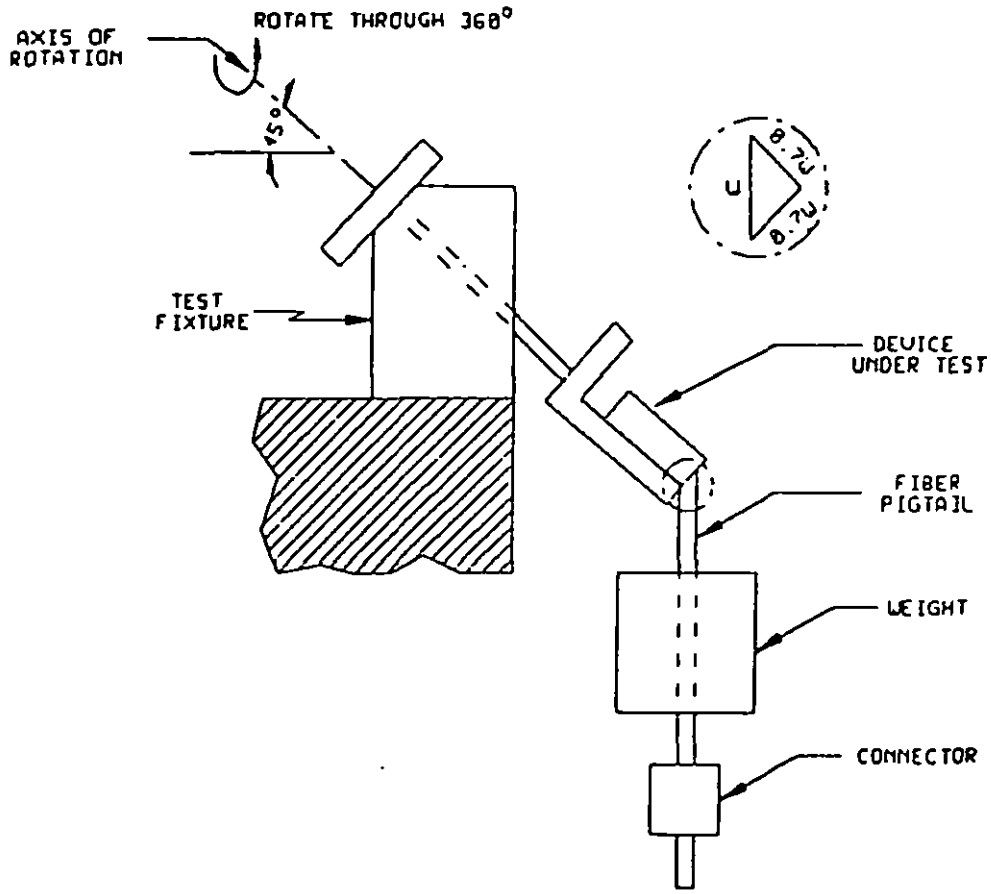


FIGURE 4. Fiber dynamic strength test set up.

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